# Digital Electronics (NECE102)

*(Decoder and Encoder)*

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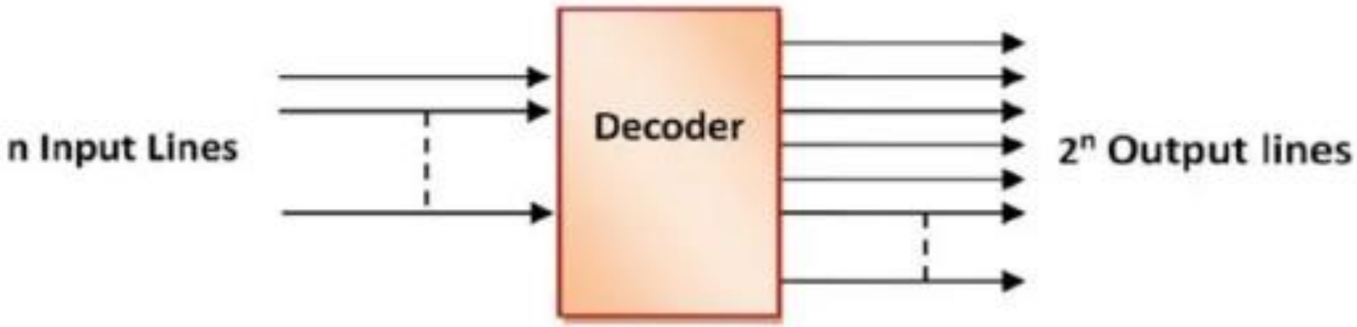
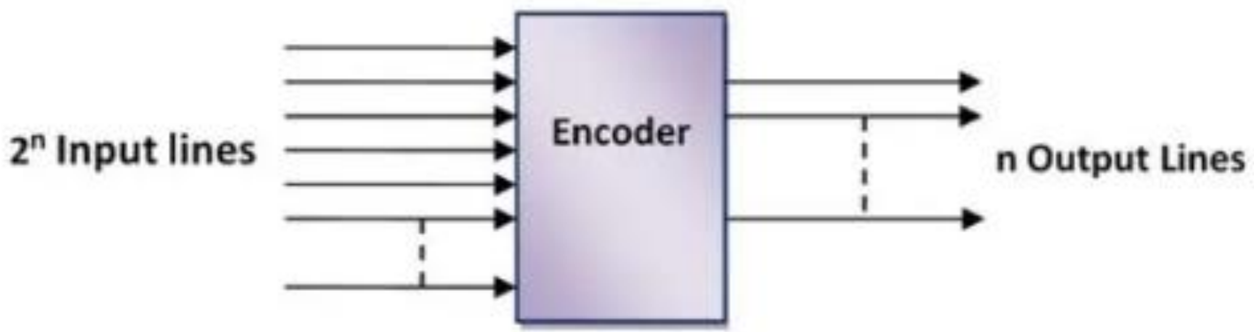
# Decoder and Encoder



# Decoder

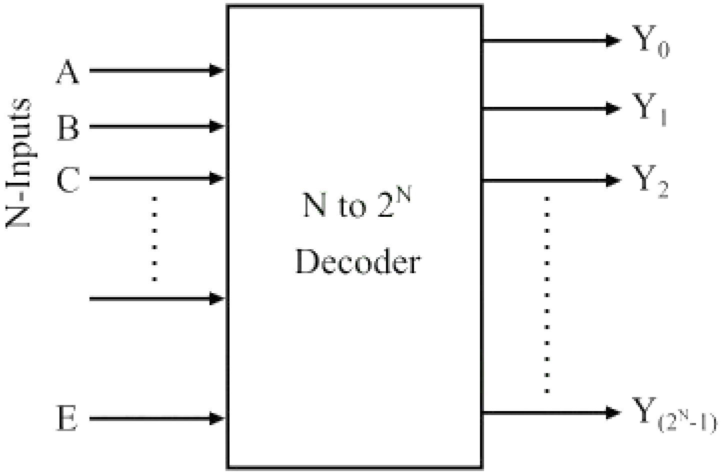
Decoder is a combinational logic circuit that converts an N-bit binary input code into M output channels in such a way that only one output channel is activated for each one of the possible combinations of inputs.

It is a multiple input and multiple output device.

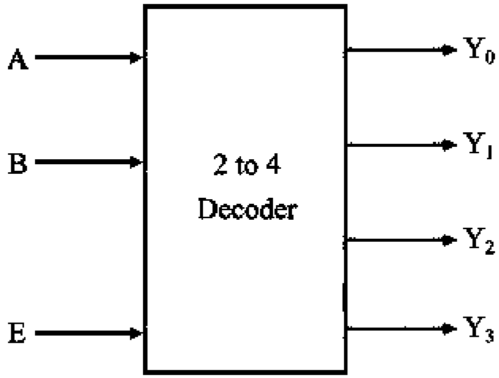
It converts N input lines into a maximum of $2^N$ output lines.

Application: to convert binary to other codes

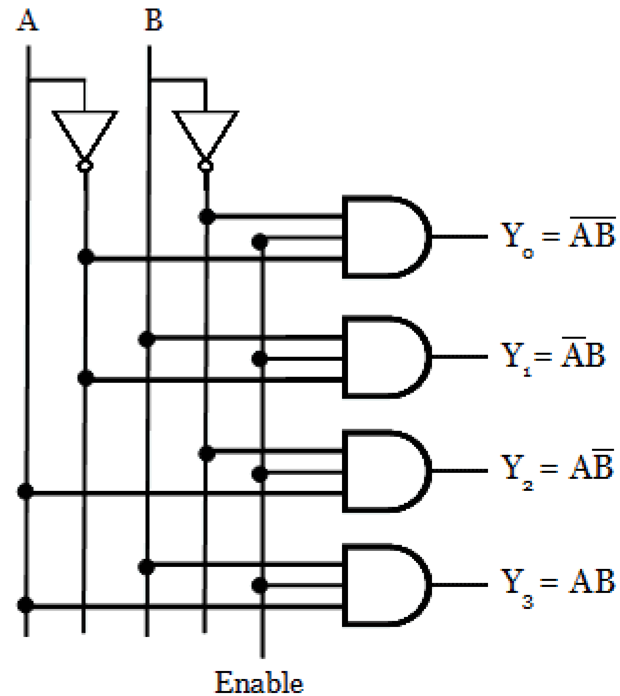
Binary to Octal
Binary to Hexadecimal
Binary to decimal



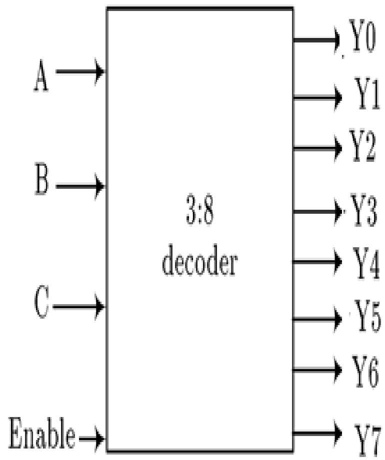
# 2 × 4 Decoder



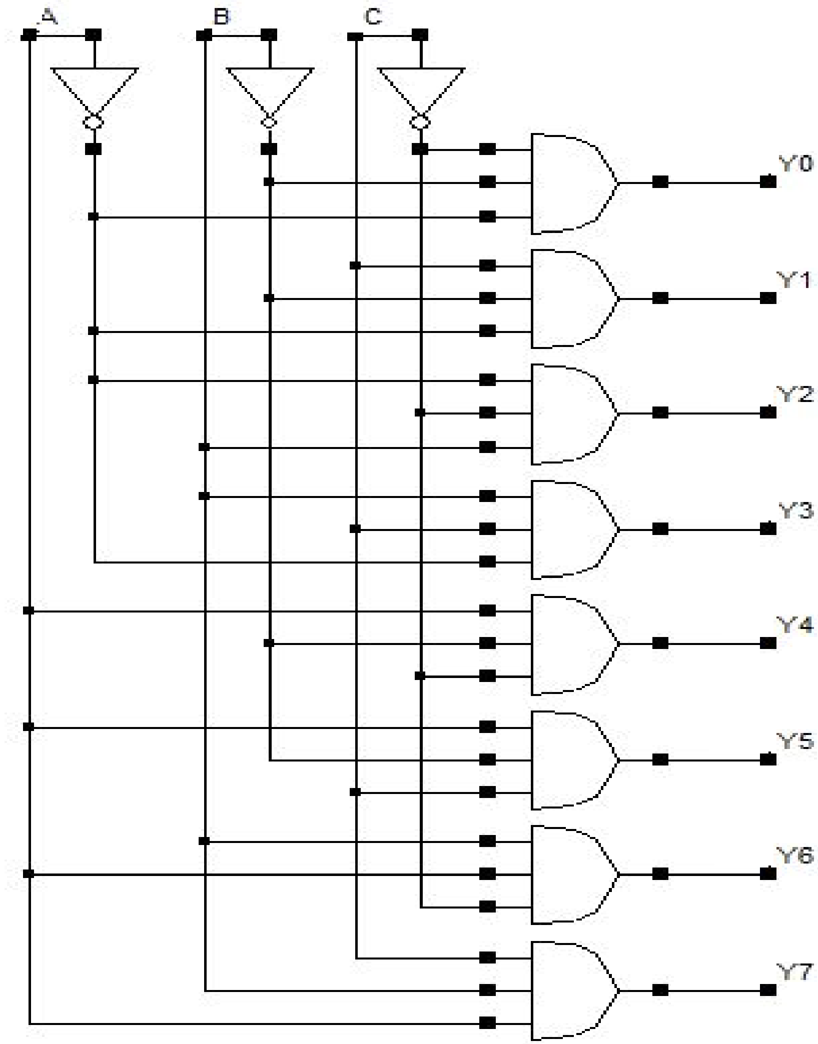

| Inputs | | | Outputs | | | |
|---|---|---|---|---|---|---|
| EN | A | B | $Y_3$ | $Y_2$ | $Y_1$ | $Y_0$ |
| 0 | × | × | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 |



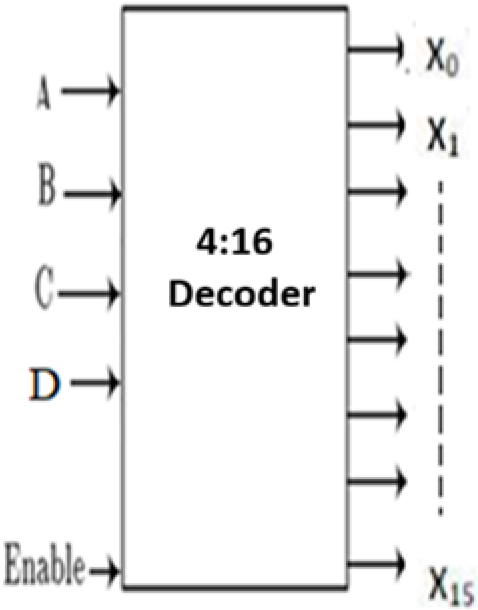
# 3 × 8 Decoder




| Inputs | | | Outputs | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | $Y_7$ | $Y_6$ | $Y_5$ | $Y_4$ | $Y_3$ | $Y_2$ | $Y_1$ | $Y_0$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

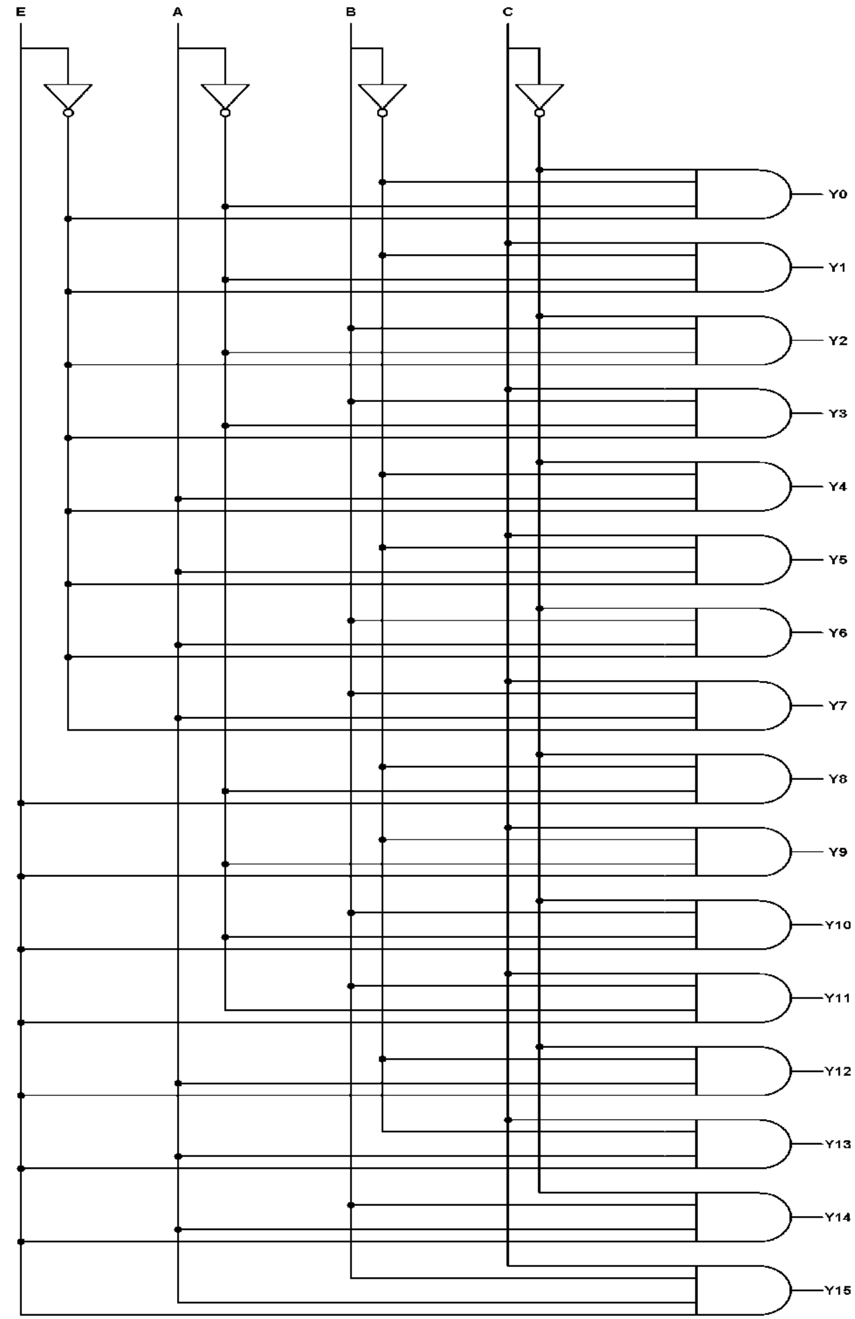


# 4 × 16 Decoder

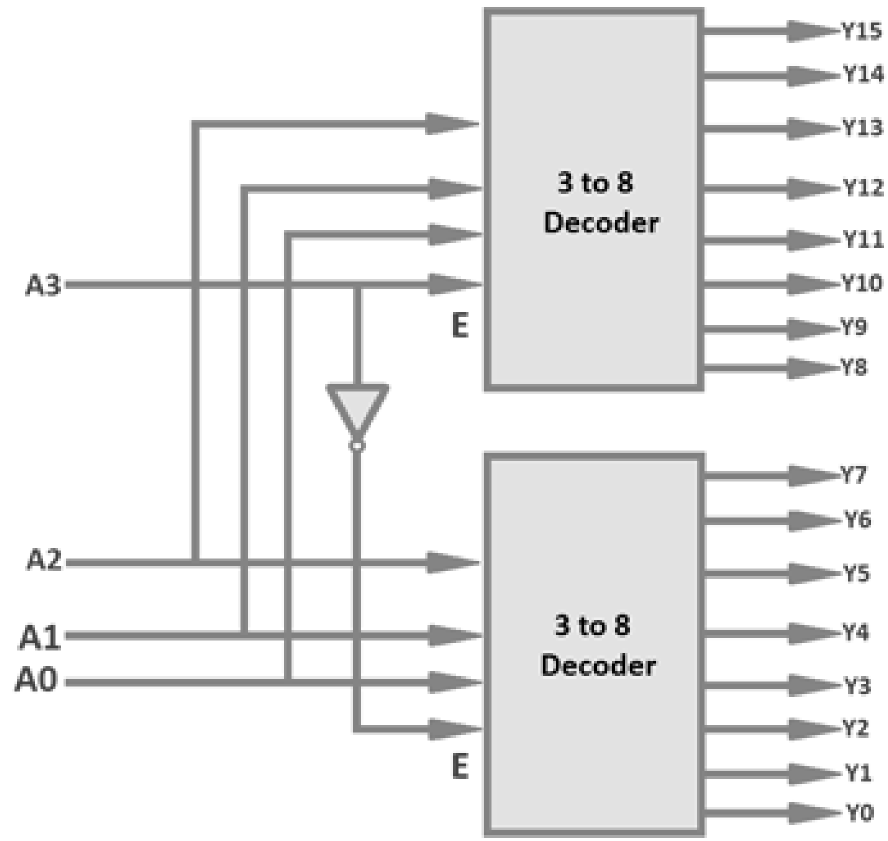



| INPUTS | | | | OUTPUTS | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | $X_0$ | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ | $X_9$ | $X_{10}$ | $X_{11}$ | $X_{12}$ | $X_{13}$ | $X_{14}$ | $X_{15}$ |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

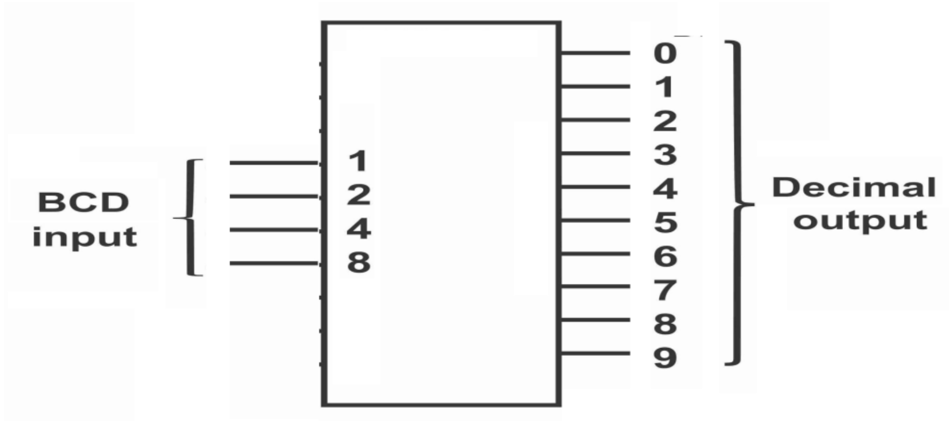
# 4 × 16 Decoder

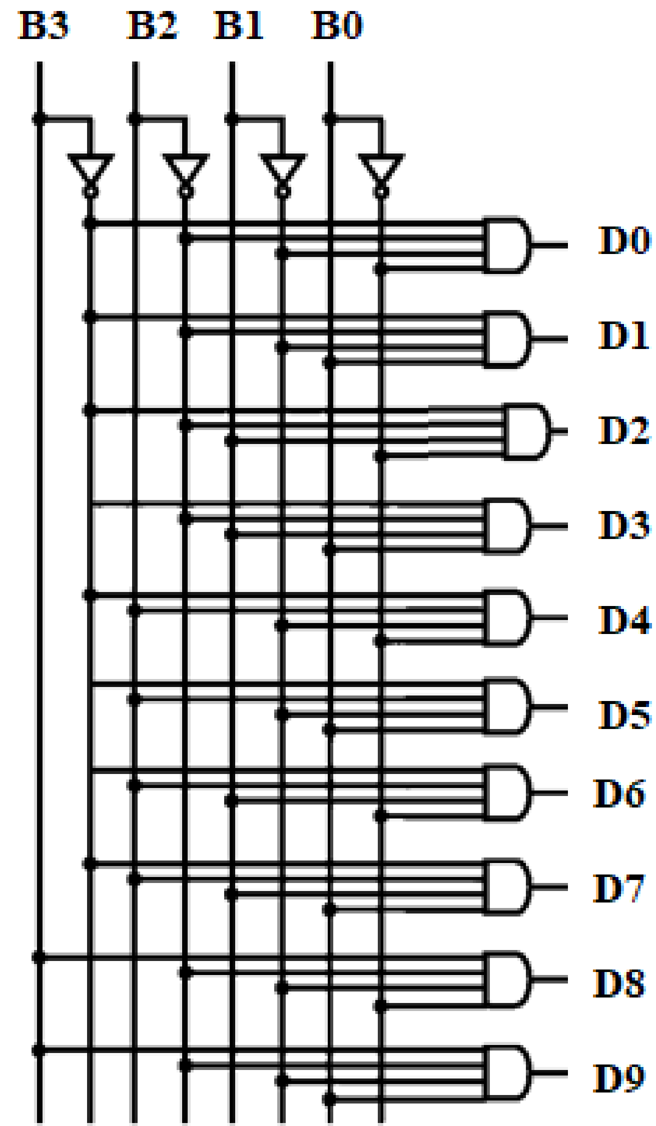


# 4 × 16 Decoder using 3 × 8 Decoder



# BCD to Decimal Decoder (4:10 decoder)



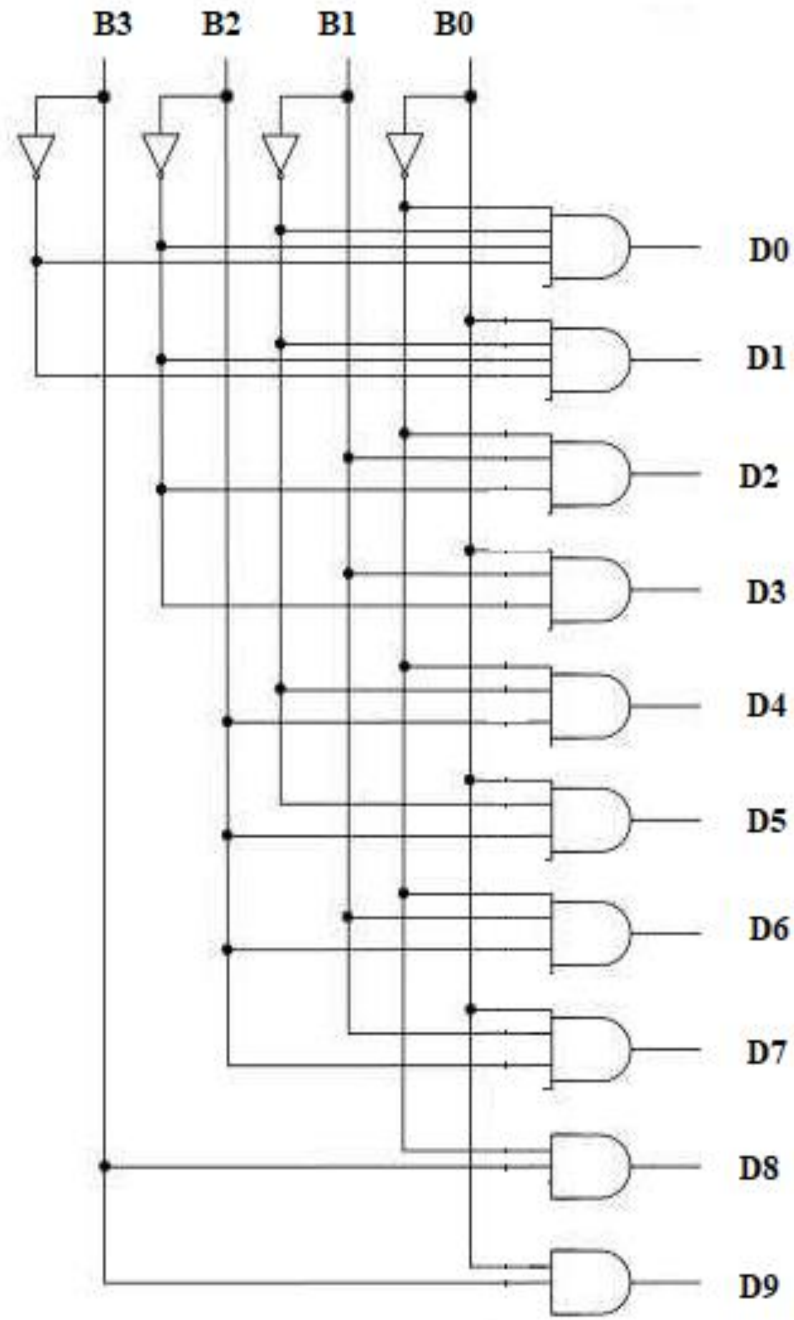

**Truth Table:**

| B3 | B2 | B1 | B0 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 | D8 | D9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

# BCD to Decimal Decoder

| B3 B2 \ B1 B0 | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | $D_0$ | $D_1$ | $D_3$ | $D_2$ |
| 01 | $D_4$ | $D_5$ | $D_7$ | $D_6$ |
| 11 | $X$ | $X$ | $X$ | $X$ |
| 10 | $D_8$ | $D_9$ | $X$ | $X$ |

**D0 = B3' B2' B1' B0'**
**D1 = B3' B2' B1' B0**
**D2 = B2' B1 B0'**
**D3 = B2' B1 B0**
**D4 = B2 B1' B0'**
**D5 = B2 B1' B0**
**D6 = B2' B1 B0'**
**D7 = B2 B1 B0**
**D8 = B3 B0'**
**D9 = B3 B0**



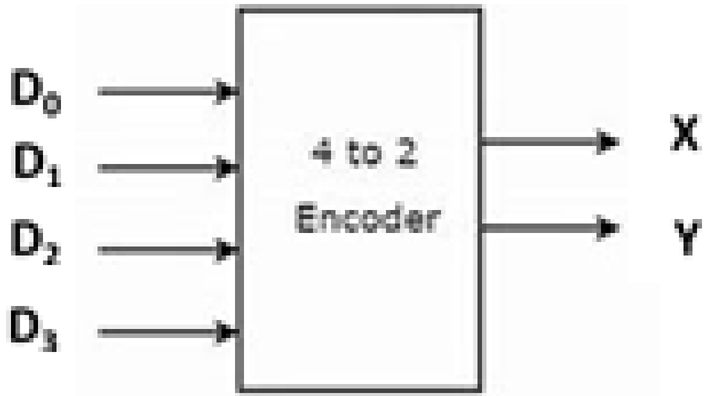
# Encoder

**Encoders –** An encoder is a combinational circuit that converts binary information in the form of a $2^N$ input lines into N output lines, which represent N bit code for the input.
For simple encoders, it is assumed that only one input line is active at a time.

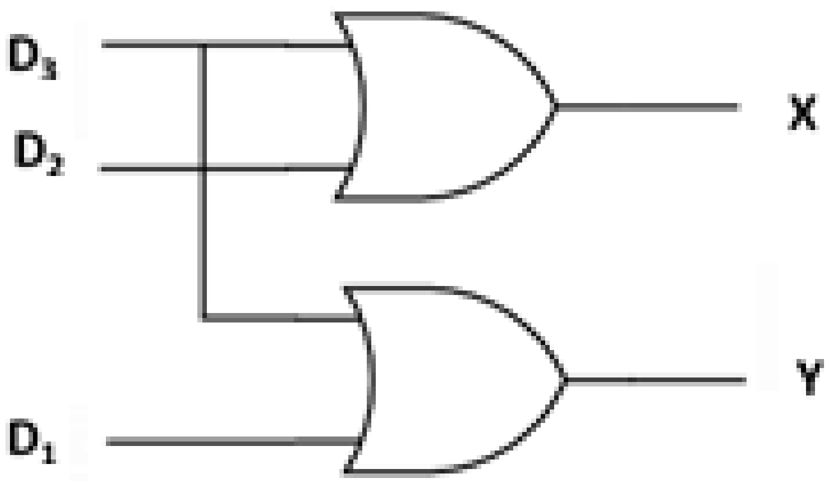
As an example, let's consider **Octal to Binary** encoder.

**Encoders** convert $2^N$ lines of input into a code of N bits and **Decoders** decode the N bits into $2^N$ lines

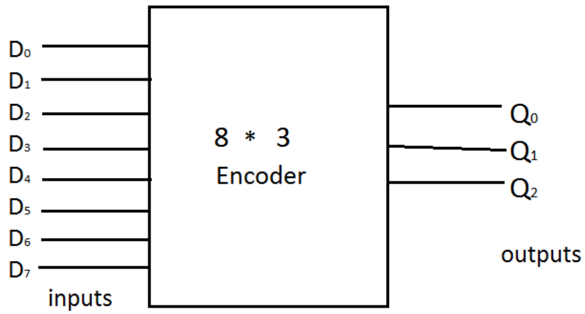
# 4 to 2 Encoder



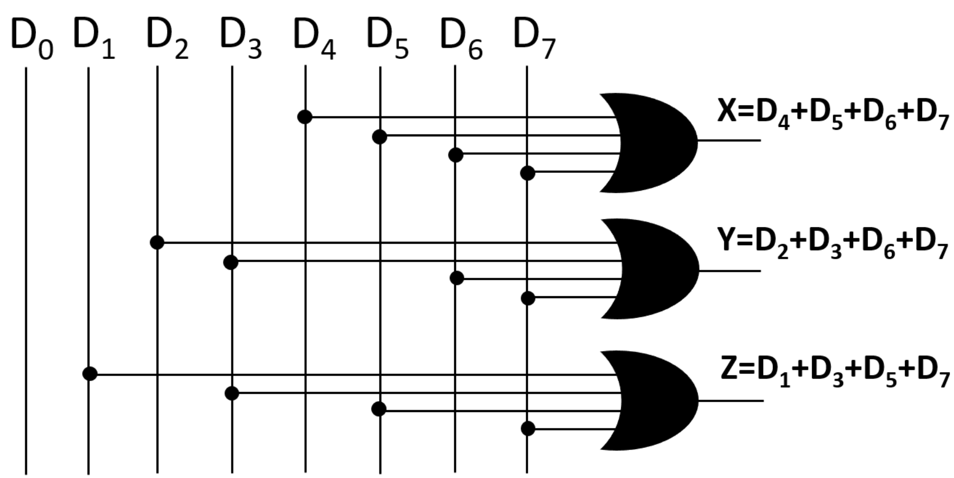

| Input | | | | Output | |
|---|---|---|---|---|---|
| $D_0$ | $D_1$ | $D_2$ | $D_3$ | X | Y |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 |



# 8 to 3 Encoder

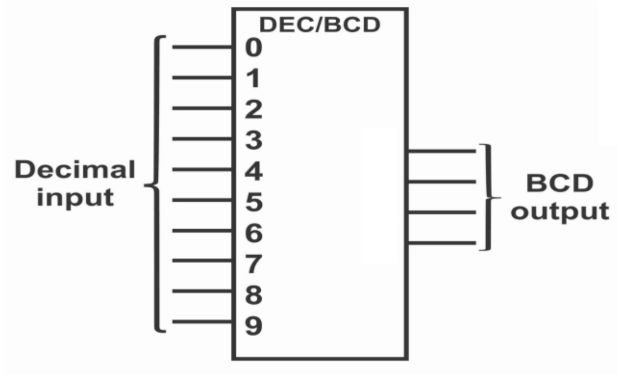


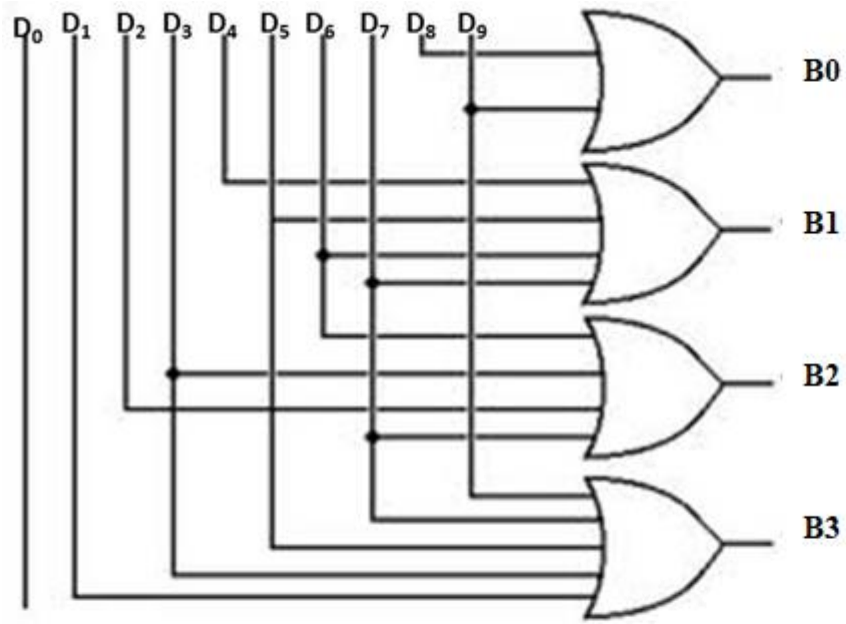

| Input | | | | | | | | Output | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $D_0$ | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | $D_7$ | X | Y | Z |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |



# Decimal to BCD Encoder

| D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | B3 | B2 | B1 | B0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |




B3 = D8 + D9
B2 = D4 + D5 + D6 + D7
B1 = D2 + D3 + D6 + D7
B0 = D1 + D3 + D5 + D7 + D9

# Digital Electronics (NECE102)

*(Code Converter)*

# Binary to Gray Code Converter

**Gray Code:**

- **It is un-weighted code**
- **It is referred as reflected binary code**
- **It is unit distance code**
- **It is cyclic code**
- **It changes in only one digit**
- **Switching time for gray code is less than any other code**

| 1 bit gray code | 2 bit gray code | 3 bit gray code | 4 bit gray code |
|---|---|---|---|
| 0 | 0 0 | 0 0 0 | 0 0 0 0 |
| 1 | 0 1 | 0 0 1 | 0 0 0 1 |
| | 1 1 | 0 1 1 | 0 0 1 1 |
| | 1 0 | 0 1 0 | 0 0 1 0 |
| | | 1 1 0 | 0 1 1 0 |
| | | 1 1 1 | 0 1 1 1 |
| | | 1 0 1 | 0 1 0 1 |
| | | 1 0 0 | 0 1 0 0 |
| | | | 1 1 0 0 |
| | | | 1 1 0 1 |
| | | | 1 1 1 1 |
| | | | 1 1 1 0 |
| | | | 1 0 1 0 |
| | | | 1 0 1 1 |
| | | | 1 0 0 1 |
| | | | 1 0 0 0 |

# Binary to Gray Code Converter (K-Map simplification)

| Decimal Number | 4 bit Binary Number $B_3 B_2 B_1 B_0$ | 4 bit Gray Code $G_3 G_2 G_1 G_0$ |
|---|---|---|
| 0 | 0 0 0 0 | 0 0 0 0 |
| 1 | 0 0 0 1 | 0 0 0 1 |
| 2 | 0 0 1 0 | 0 0 1 1 |
| 3 | 0 0 1 1 | 0 0 1 0 |
| 4 | 0 1 0 0 | 0 1 1 0 |
| 5 | 0 1 0 1 | 0 1 1 1 |
| 6 | 0 1 1 0 | 0 1 0 1 |
| 7 | 0 1 1 1 | 0 1 0 0 |
| 8 | 1 0 0 0 | 1 1 0 0 |
| 9 | 1 0 0 1 | 1 1 0 1 |
| 10 | 1 0 1 0 | 1 1 1 1 |
| 11 | 1 0 1 1 | 1 1 1 0 |
| 12 | 1 1 0 0 | 1 0 1 0 |
| 13 | 1 1 0 1 | 1 0 1 1 |
| 14 | 1 1 1 0 | 1 0 0 1 |
| 15 | 1 1 1 1 | 1 0 0 0 |

| $b_3b_2$ \ $b_1b_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 0 | 0 | 0 | 0 |
| 11 | 1 | 1 | 1 | 1 |
| 10 | 1 | 1 | 1 | 1 |

$$g_3 = b_3$$

| $b_3b_2$ \ $b_1b_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 1 | 1 | 1 | 1 |
| 11 | 0 | 0 | 0 | 0 |
| 10 | 1 | 1 | 1 | 1 |

$$g_2 = \bar{b}_2 b_3 + b_2 \bar{b}_3 = b_2 \oplus b_3$$

# Binary to Gray Code Converter (K-Map simplification)

| Decimal Number | 4 bit Binary Number $B_3 B_2 B_1 B_0$ | 4 bit Gray Code $G_3 G_2 G_1 G_0$ |
|---|---|---|
| 0 | 0 0 0 0 | 0 0 0 0 |
| 1 | 0 0 0 1 | 0 0 0 1 |
| 2 | 0 0 1 0 | 0 0 1 1 |
| 3 | 0 0 1 1 | 0 0 1 0 |
| 4 | 0 1 0 0 | 0 1 1 0 |
| 5 | 0 1 0 1 | 0 1 1 1 |
| 6 | 0 1 1 0 | 0 1 0 1 |
| 7 | 0 1 1 1 | 0 1 0 0 |
| 8 | 1 0 0 0 | 1 1 0 0 |
| 9 | 1 0 0 1 | 1 1 0 1 |
| 10 | 1 0 1 0 | 1 1 1 1 |
| 11 | 1 0 1 1 | 1 1 1 0 |
| 12 | 1 1 0 0 | 1 0 1 0 |
| 13 | 1 1 0 1 | 1 0 1 1 |
| 14 | 1 1 1 0 | 1 0 0 1 |
| 15 | 1 1 1 1 | 1 0 0 0 |

| | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 3 | 2 |
| 01 | 4 | 5 | 7 | 6 |
| 11 | 12 | 13 | 15 | 14 |
| 10 | 8 | 9 | 11 | 10 |

# Binary to Gray Code Converter

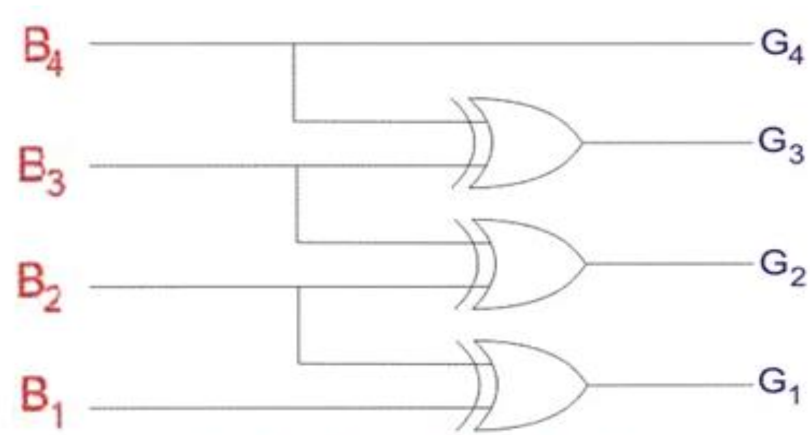
| Decimal Number | 4 bit Binary Number $B_3 B_2 B_1 B_0$ | 4 bit Gray Code $G_3 G_2 G_1 G_0$ |
|---|---|---|
| 0 | 0 0 0 0 | 0 0 0 0 |
| 1 | 0 0 0 1 | 0 0 0 1 |
| 2 | 0 0 1 0 | 0 0 1 1 |
| 3 | 0 0 1 1 | 0 0 1 0 |
| 4 | 0 1 0 0 | 0 1 1 0 |
| 5 | 0 1 0 1 | 0 1 1 1 |
| 6 | 0 1 1 0 | 0 1 0 1 |
| 7 | 0 1 1 1 | 0 1 0 0 |
| 8 | 1 0 0 0 | 1 1 0 0 |
| 9 | 1 0 0 1 | 1 1 0 1 |
| 10 | 1 0 1 0 | 1 1 1 1 |
| 11 | 1 0 1 1 | 1 1 1 0 |
| 12 | 1 1 0 0 | 1 0 1 0 |
| 13 | 1 1 0 1 | 1 0 1 1 |
| 14 | 1 1 1 0 | 1 0 0 1 |
| 15 | 1 1 1 1 | 1 0 0 0 |

| $b_3b_2$ \ $b_1b_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 1 | 1 |
| 01 | 1 | 1 | 0 | 0 |
| 11 | 1 | 1 | 0 | 0 |
| 10 | 0 | 0 | 1 | 1 |

$$g_1 = \bar{b}_2 b_1 + b_2 \bar{b}_1 = b_1 \oplus b_2$$

| $b_3b_2$ \ $b_1b_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 0 | 1 |
| 01 | 0 | 1 | 0 | 1 |
| 11 | 0 | 1 | 0 | 1 |
| 10 | 0 | 1 | 0 | 1 |

$$g_0 = \bar{b}_1 b_0 + b_1 \bar{b}_0 = b_1 \oplus b_0$$



Logic Circuit for Binary to Gray Code Converter

# Gray to Binary Code Converter

| | 4 bit Gray Code $G_3G_2G_1G_0$ | 4 bit Binary Code $B_3B_2B_1B_0$ |
|---|---|---|
| 0 | 0 0 0 0 | 0 0 0 0 |
| 1 | 0 0 0 1 | 0 0 0 1 |
| 3 | 0 0 1 1 | 0 0 1 0 |
| 2 | 0 0 1 0 | 0 0 1 1 |
| 6 | 0 1 1 0 | 0 1 0 0 |
| 7 | 0 1 1 1 | 0 1 0 1 |
| 5 | 0 1 0 1 | 0 1 1 0 |
| 4 | 0 1 0 0 | 0 1 1 1 |
| 12 | 1 1 0 0 | 1 0 0 0 |
| 13 | 1 1 0 1 | 1 0 0 1 |
| 15 | 1 1 1 1 | 1 0 1 0 |
| 14 | 1 1 1 0 | 1 0 1 1 |
| 10 | 1 0 1 0 | 1 1 0 0 |
| 11 | 1 0 1 1 | 1 1 0 1 |
| 9 | 1 0 0 1 | 1 1 1 0 |
| 8 | 1 0 0 0 | 1 1 1 1 |

K-map for B3:

| G3G2 \ G1G0 | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 0 | 0 | 0 | 0 |
| 11 | 1 | 1 | 1 | 1 |
| 10 | 1 | 1 | 1 | 1 |

B3 = G3

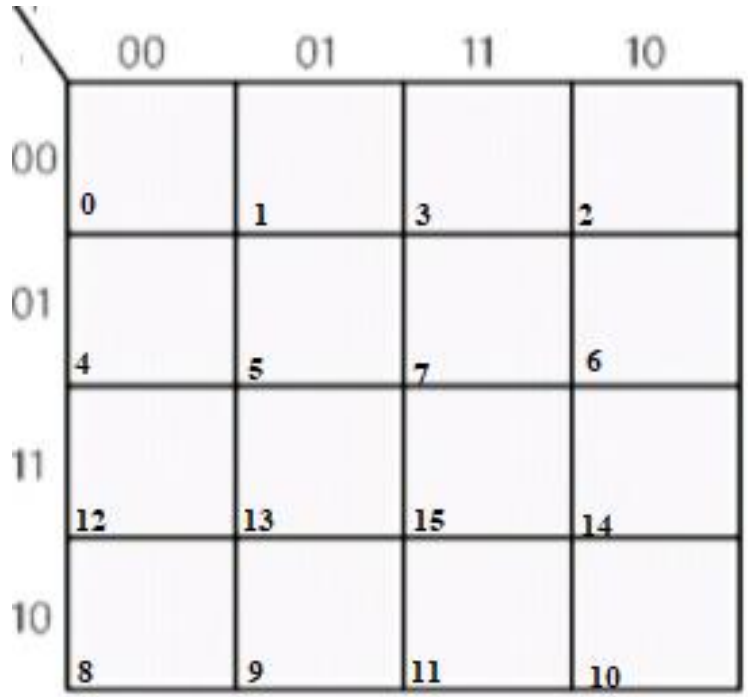
K-map for B2:

| G3G2 \ G1GO | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 1 | 1 | 1 | 1 |
| 11 | 0 | 0 | 0 | 0 |
| 10 | 1 | 1 | 1 | 1 |

$B2 = G3 \oplus G2$

# Gray to Binary Code Converter

| | 4 bit Gray Code $G_3 G_2 G_1 G_0$ | 4 bit Binary Code $B_3 B_2 B_1 B_0$ |
|---|---|---|
| 0 | 0 0 0 0 | 0 0 0 0 |
| 1 | 0 0 0 1 | 0 0 0 1 |
| 3 | 0 0 1 1 | 0 0 1 0 |
| 2 | 0 0 1 0 | 0 0 1 1 |
| 6 | 0 1 1 0 | 0 1 0 0 |
| 7 | 0 1 1 1 | 0 1 0 1 |
| 5 | 0 1 0 1 | 0 1 1 0 |
| 4 | 0 1 0 0 | 0 1 1 1 |
| 12 | 1 1 0 0 | 1 0 0 0 |
| 13 | 1 1 0 1 | 1 0 0 1 |
| 15 | 1 1 1 1 | 1 0 1 0 |
| 14 | 1 1 1 0 | 1 0 1 1 |
| 10 | 1 0 1 0 | 1 1 0 0 |
| 11 | 1 0 1 1 | 1 1 0 1 |
| 9 | 1 0 0 1 | 1 1 1 0 |
| 8 | 1 0 0 0 | 1 1 1 1 |



# Gray to Binary Code Converter

| | 4 bit Gray Code $G_3 G_2 G_1 G_0$ | 4 bit Binary Code $B_3 B_2 B_1 B_0$ |
|---|---|---|
| 0 | 0 0 0 0 | 0 0 0 0 |
| 1 | 0 0 0 1 | 0 0 0 1 |
| 3 | 0 0 1 1 | 0 0 1 0 |
| 2 | 0 0 1 0 | 0 0 1 1 |
| 6 | 0 1 1 0 | 0 1 0 0 |
| 7 | 0 1 1 1 | 0 1 0 1 |
| 5 | 0 1 0 1 | 0 1 1 0 |
| 4 | 0 1 0 0 | 0 1 1 1 |
| 12 | 1 1 0 0 | 1 0 0 0 |
| 13 | 1 1 0 1 | 1 0 0 1 |
| 15 | 1 1 1 1 | 1 0 1 0 |
| 14 | 1 1 1 0 | 1 0 1 1 |
| 10 | 1 0 1 0 | 1 1 0 0 |
| 11 | 1 0 1 1 | 1 1 0 1 |
| 9 | 1 0 0 1 | 1 1 1 0 |
| 8 | 1 0 0 0 | 1 1 1 1 |

K-map for B1:

| G3G2 \ G1G0 | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 1 | 1 |
| 01 | 1 | 1 | 0 | 0 |
| 11 | 0 | 0 | 1 | 1 |
| 10 | 1 | 1 | 0 | 0 |

$B1 = G3 \oplus G2 \oplus G1$

K-map for B0:

| G3G2 \ G1G0 | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 0 | 1 |
| 01 | 1 | 0 | 1 | 0 |
| 11 | 0 | 1 | 0 | 1 |
| 10 | 1 | 0 | 1 | 0 |

$BO = G3 \oplus G2 \oplus G1 \oplus G0$

# BCD to Excess 3 Code Converter

| Decimal Number | BCD code (Input) | | | | Excess-3 code (Output) | | | |
|---|---|---|---|---|---|---|---|---|
| | $D_3$ | $D_2$ | $D_1$ | $D_0$ | $E_3$ | $E_2$ | $E_1$ | $E_0$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 5 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 6 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 7 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 8 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 9 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

$$E_3 = \sum m(5, 6, 7, 8, 9) + d\,(10, 11, 12, 13, 14, 15)$$

For $E_3$ output

| $D_3D_2$ \ $D_1D_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 0 | 1 | 1 | 1 |
| 11 | X | X | X | X |
| 10 | 1 | 1 | X | X |

$$E_3 = D_3 + D_2D_0 + D_2D_1$$

$$E_2 = \sum m(1, 2, 3, 4, 9) + d\,(10, 11, 12, 13, 14, 15)$$

For $E_2$ output

| $D_3D_2$ \ $D_1D_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 1 | 1 |
| 01 | 1 | 0 | 0 | 0 |
| 11 | X | X | X | X |
| 10 | 0 | 1 | X | X |

$$E_2 = D_2\overline{D}_1\overline{D}_0 + \overline{D}_2D_0 + \overline{D}_2D_1$$

# BCD to Excess 3 Code Converter

| Decimal Number | BCD code (Input) | | | | Excess-3 code (Output) | | | |
|---|---|---|---|---|---|---|---|---|
| | $D_3$ | $D_2$ | $D_1$ | $D_0$ | $E_3$ | $E_2$ | $E_1$ | $E_0$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 5 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 6 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 7 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 8 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 9 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

$$E_1 = \sum m(0, 3, 4, 7, 8) + d\,(10, 11, 12, 13, 14, 15)$$

For $E_1$ output

| $D_3D_2 \backslash D_1D_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 1 | 0 |
| 01 | 1 | 0 | 1 | 0 |
| 11 | x | x | x | x |
| 10 | 1 | 0 | x | x |

$$E_1 = \overline{D}_1\overline{D}_0 + D_1D_0$$

$$E_0 = \sum m(0, 2, 4, 6, 8,) + d\,(10, 11, 12, 13, 14, 15)$$

For $E_0$ output

| $D_3D_2 \backslash D_1D_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 0 | 1 |
| 01 | 1 | 0 | 0 | 1 |
| 11 | x | x | x | x |
| 10 | 1 | 0 | x | x |

$$E_0 = \overline{D}_0$$

| Decimal Number | BCD code (Input) | | | | Excess-3 code (Output) | | | |
|---|---|---|---|---|---|---|---|---|
| | D3 | D2 | D1 | Do | E3 | E2 | E1 | Eo |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 5 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 6 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 7 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 8 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 9 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

$E_3 = \sum m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$

$E_2 = \sum m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$

$E_1 = \sum m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$

$E_0 = \sum m(0, 2, 4, 6, 8,) + d(10, 11, 12, 13, 14, 15)$

| | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 3 | 2 |
| 01 | 4 | 5 | 7 | 6 |
| 11 | 12 | 13 | 15 | 14 |
| 10 | 8 | 9 | 11 | 10 |

# BCD to Excess 3 Code Converter

$$E_3 = D_3 + D_2D_0 + D_2D_1$$

$$E_2 = D_2\overline{D}_1\overline{D}_0 + \overline{D}_2D_0 + \overline{D}_2D_1$$

$$E_1 = \overline{D}_1\overline{D}_0 + D_1D_0$$

$$E_0 = \overline{D}_0$$